

Mod SDK changelog

This is the changelog for the Core Keeper Mod SDK. It will contain changes to the SDK itself as well as internal changes to the project that isn't covered in the regular changelog.

It is still very much recommended to also read what is in the changelog for the Core Keeper application on Steam: <https://store.steampowered.com/news/app/1621690?updates=true>

1.2.1.0

- CraftingAuthoring now uses a `ModdedObjectID` string when ObjectID is set to ObjectID.None, which'll be converted to an ObjectID during runtime. You may use this change to add modded items to workbenches and such.
- All Graphical Prefabs which have a component that inherits from IPoolable will now automatically be pooled. In order to adjust the size of pooling or to disable it you may create a `PooledModGraphicalObjectBank` Scriptable Object via the Create menu.
- The Sprite Render on the Mouse cursor has been upgraded to a SpriteAsset which can now be modded via the ScriptableDataBlock editor.
- QuantumConsole has now been exposed to mod developers and can be switched on by setting MenuManager.enableConsole to true. It can then be opened in-game via ctrl+x. The object suggestor supports modded Object Names when using the spawnByName command.

1.2.0.3

- More comprehensive mod SDK documentation is now available at <https://modding.corekeepergame.com/>.
- UnityExtensions has moved to a new PugUnityExtensions package and Pug.UnityExtensions namespace.
- PugProperties has moved to a new PugProperties package and Pug.Properties namespace.
- TileMapLookup.GetBlockingTile and TileMapLookup.HasBlockingTile has been deprecated in favor of TileMapLookup.TryGetBlockingTile and will get removed in a future release.
- PugAutomation namespace has been renamed to Pug.Automation.
- PugElectricity namespace has been merged into Pug.Automation.
- ElectricityCD is now part of the Pug.Automation namespace.
- PugConversion has been extracted as a separate package. There are a few changes that may require you to update any existing converter code.
 - PugConversion namespace has been renamed to Pug.Conversion.
 - PugConverter and PugPostConverter have been renamed to Converter and PostConverter, respectively.
 - The ConditionsTable asset is no longer cached on each individual converter. Converters that need access to the conditions table should load and cache it themselves. See e.g. LevelEntitiesBufferConverter for an example.

- Added BurstDisabler API to selectively disable burst compiler for specific systems. This is currently limited to disabling Burst for ISystem. We are working on extending this to work with Burst-compiled jobs as well. You may see some jobs running without Burst using this API currently, but results are inconsistent.
- Core Keeper now uses ScriptableDataBlocks which are just like Scriptable Objects but hold an address synced to the assets' GUID.
- Gradient Maps are now stored as a texture array inside of GradientMapDataBlocks.
- SpriteAssets are now SpriteAssetDataBlocks.
- SpriteAssetSkins are now SpriteAssetSkinDataBlocks.
- PlayerCustomizationTable is now PlayerCustomizationTableDataBlock.
- Sprite Objects now reference SpriteAssetDataBlocks and GradientMapDataBlocks.
- Mods can now be loaded from Steam Workshop as well.
- Localization now works through the TextDataBlock system, to add or change in-game text you will need to create a new TextDataBlock instead of a Localization.csv file.
- A majority of the ModSDK has been moved from assets into a PugMod package.

1.1.2.8

- Upgraded the Unity engine version from 6000.0.58f2 to 6000.0.59f2. This was required to fix a frequent source of crashes, and it is not expected to impact modding. You can download the new version in the Unity 6 > LTS section in the download archive: <https://unity.com/releases/editor/archive>. If you are having issues after upgrading, try closing the editor and delete the project's Library folder before reopening.

1.1.2.6

- Upgraded the Unity engine version from 2022.3.62f2 to 6000.0.58f2. You can download the new version in the Unity 6 > LTS section in the download archive: <https://unity.com/releases/editor/archive>. If you are having issues after upgrading, try closing the editor and delete the project's Library folder before reopening.
- Removed the obsolete Visual Studio Code Editor package. The Visual Studio Editor package now includes support for VS Code.
- Removed the OpenProject scripts. Instead, you should open the project using the [Unity Hub](#) available for download on Unity's website.

1.1.2.5

- Upgraded the Unity engine version from 2022.3.50f1 to 2022.3.62f2. You can download the new version in the Unity 2022 > LTS section in [the download archive](#). If you are having issues after upgrading, try closing the editor and delete the project's Library folder before reopening. This upgrade is motivated by the security vulnerability found in older versions of the Unity engine. For more information, see [Unity's official announcement](#).
- Addressables is now included as an embedded package in the SDK.

1.1.2.0

- `UnityExtensions.ExtensionMethods.RoundToMultiple(x, m)` now correctly rounds x to the nearest multiple of m as the name indicates, i.e. it returns $k*m$ for an integer k such that $k*m + r = x$ and $-m / 2 \leq r < m / 2$. It used to instead round to the reciprocal $1/m$. This change means all existing code that calls these methods must be adjusted.
- Improved performance of `AlphabeticallySortedEnumPropertyDrawer`. Thank you Limoka for this suggestion!
- `SpriteAssetSkin` now has a project view preview like `SpriteAsset` already did.
- `SpriteAssets` now support up to three separate gradient maps. The “grayscale” textures have been generalized so that the red channel selects a color from the primary gradient map, the green from the secondary, and the blue from the tertiary. This allows separate parts of the sprite to use different color maps.
- The `SpriteAsset` and `SpriteAssetSkin` gradient maps have changed due to the above change. If you used the `SpriteAsset[Skin]` inspector to generate the grayscale and gradient map textures, you can update the gradient format using the “Unmap to grayscale” option.
- Improved `PickStringFromEnumDrawer` to allow fuzzy searching of base game object IDs. Thank you Limoka for this contribution!
- Fixed some issues when using SDK on a Linux machine. Thank you Barroit for these contributions!

1.1.0

- Removed all fields related to sprite objects in `EntityMonoBehavior.reskinOptions` in favor of dedicated components like `SpriteSkinFromEntityAndSeason` and `SpriteMaterialFromEntity`. These fields have been deprecated since 0.7.3 and are no longer used by any non-modded objects.
- Renamed the `SpriteInstancing` package and namespace to `Pug.Sprite`. This breaks visuals in existing mods which use `SpriteObject/SpriteAsset`. Rebuilding the mods in the updated Mod SDK project should fix all references automatically.
- Renamed the `PugRP` package and namespace to `Pug.RP`.
- Include all `System.*` assemblies when updating the SDK from game files. Previously especially `System.Collections.Immutable` was missing causing some errors with `InteractableObject` script.
- Added modified DOTS packages embedded in the SDK project (can be found under `Packages/`). These are still fairly close to the public package versions in what they expose publicly but there have been more and more minor changes so we are including all of them for clarity and to simplify the SDK update process.

1.0.1.11

- Integrated some QoL life changes from pull request from Limoka:
<https://github.com/Pugstorm/CoreKeeperModSDK/pull/7>

1.0.1.7

- Updated Unity version from 2022.3.20 to 2022.3.50. Unity Hub should prompt you to install the updated version next time you open the project. The Unity upgrade mainly contains engine bugfixes, so there are no expected changes in mod behavior.

1.0.1

- The following components have been merged into one component named DropLootAuthoring:
 - DropsLootAuthoring
 - DropsLootFromLootTableAuthoring
 - SeasonalLootAuthoring
 - DropsLootOnUseAuthoring
 - DropsLootWhenDamagedAuthoring
- HealthRegenerationAuthoring has been merged into HealthAuthoring.
- AnimationOrientationAuthoring has been merged into AnimationAuthoring.
- GrowingAuthoring options has been merged into SeedAuthoring and PlantAuthoring.
- LastAttackerAuthoring has been merged into EnemyAuthoring.
- Removed canBeHitAtSameTimeAsEnemy bool in MineableAuthoring as it was only used by PlayerGrave.
- Chests no longer need to be adjacent to the crafting station to pick materials from the chests but can be anywhere within a 10 tiles radius of the crafting station. The code that handles this has been reworked to function very differently from before.
- The ClientInput component isn't set outside of the prediction loop anymore. If you need to fetch or modify some input value outside of the prediction loop, fetch the data from ClientInputData. See updated InvertPlayerMovementSystem in SystemExample.

1.0

- Core Keeper is now available on consoles as well as PC. Modding is only available on PC, and we will most likely never support mods on consoles in the same way we do on PC. The platform owners (i.e. Sony, Microsoft and Nintendo) have strict requirements for all games that are released on their consoles, and there is no reasonable way to comply with them if players can modify the game to any meaningful extent.
- There is a known issue with mods not working properly on non-steam platforms due to some differences in the game initialization flow. We are working on a fix for this.
- Player controller has been changed to be server authoritative instead. This was also mentioned in the game changelog, but this is worth bringing up here as well as one of the biggest refactorings we've done. This means the player controller consists of a number of ECS systems. The code is still similar in places, but for example PlayerController have more static functions instead called from ECS.
- Apart from the player most enemies and objects are interpolated (shown with a server delay to avoid client-side computation). Some player interactions can still

make them be predicted instead to improve responsiveness which means that a bunch of logic are now supporting prediction meaning a lot more systems are designed to run both on server and client.

- Completely new interaction system which is triggered from ECS. This part had to be completely remade instead of ported. Callbacks on EntityMonoBehaviours like OnUse should still be called in the same way via the new system.
- All SinglePugMap methods have been refactored to use world coordinates instead of render coordinates. Use EntityMonoBehaviour.ToWorldFromRender or Manager.camera.RenderOrigo to convert from render to world coordinates.
- RenderOrigo has been renamed to the more accurate RenderAnchor. You can access these using Manager.camera.GetRenderAnchor. If you don't need it for the full session, call Manager.camera.ReturnRenderAnchor to return it so that it can be cleaned up.
- PugSprite has been completely removed. This component was used to create MeshRenderers from SpriteRenderer to get the batching support from Unity, but has since been replaced mostly by using SpriteObject instead of SpriteRenderer.
- Even more enemies and objects have been converted from SpriteRenderer to using our own sprite rendering solution SpriteObject.
- More parts of the code is now using Burst, especially after the player controller rework. This is mainly done to improve performance. We are aware that Burst-compiled code is a lot harder to patch, and we are already looking into ways to make this easier for modding.
- The new world generation uses a new GPU-driven framework to generate the terrain. This data is processed by an ECS system to create the ECS data that describes a world. The rules for this processing are described in the `TileTypeMapping` asset. You can also get direct access to the data produced by world generation shaders via the helper struct created by `Manager.worldgen.CreateProceduralDataRequester`. Each position in-game is described by a RGBA pixel. The current format is as follows:
 - Red channel: encodes the tile type. Use `CoreKeeperWorld.GetTileType` to get a more meaningful enum value.
 - Green channel: encodes the biome. Use `CoreKeeperWorld.GetBiome` to get a more meaningful enum value.
 - Blue channel: bit 0 (least significant) encodes if the position has only floor and no wall, bit 1 if there is a roof hole, and bit 3 if the Great Wall occupies this position. All other bits are currently unused.
 - Alpha channel: bits 0-2 encodes the ore type at this position. All other bits are currently unused.
- SerializeWorldSystem has been reworked to only load in things to the "live" world when the player is close enough (200 tiles). Previously, entities outside this range would be disabled but would still be kept in the live world. Some entities are always loaded like electricity, automation objects and bosses. The unloaded objects are kept in a separate ECS world (SerializeWorld) with a minimal representation. This world is what the world save file contains.
- AnimationAuthoring script was moved inadvertently, if you used this in any custom GameObject, you'll have to reassign it. We try to not break references like that, but this one slipped the cracks in some refactor.

- WindUpAuthoring removed and same settings are moved to SecondaryUseAuthoring.
- Custom EntityMonoBehaviours aren't updated as automatically as before, you might have to call some functions manually via GraphicalUpdate (see EnemyExample).
- Addressables are used in various places to load and unload assets to minimize memory footprint.
- A bunch of TimerSimple fields have been changed to use TickTimer. The typical reason is that this timer needs to be synchronized between server and client meaning that it needs to use NetworkTick instead of a regular floating point value to measure time.

This update contains lots of other small internal changes. Too many to list all of them here, but feel free to ask in the Discord and we'll do our best to answer your questions.

0.7.5

- Updated Unity version to 2022.3.20f1. This is mostly in preparation for 1.0 and while it means there will be some changes we don't expect anything noticeable but you will need the new version for the Mod SDK as well.
- For performance reasons, the update frequency of the ECS server world has been decreased from 30 to 20 updates per second. Any modded ECS code that uses the number of elapsed ticks as a metric for when to update will consequently run slower and may need to be adjusted. Similarly, MonoBehaviour.FixedUpdate is now called 30 times per second, down from 60 previously. Client world systems, objects implementing IGraphicalObject, and MonoBehaviour that only call the non-fixed update methods are unaffected, as they are all still called once per frame.
- All EntityUtility methods that previously accepted a ComponentType parameter have been refactored to instead take the relevant IComponentData type as a generic type parameter. Previously, each call to these functions resulted in a C# reflection lookup, which accumulated to have a significant performance impact.

0.7.4

- The mods that are loaded from mod.io will have an extra check that verifies the current version is supported by the mod, indicated by the Game Version tag. If this isn't set by the mod to the current version, it will still be possible to load the mod, but the player will get a warning first. This means it is more important to set this version tag correctly than before.
- The HealthChangeBuffer now has an entity field which indicates which entity is affected by the change and is stored in a singleton instead of on each entity.
- Expanded usage of the PugProperties system to avoid storing as much data on entities which instead can be shared. Each object entity now has an ObjectPropertiesCD component that can be used to look up shared data for that object. Added SetProperty functions to PugConverter where this data can be set.

- Some of the tile functions in `Manager.multiMap` have been replaced with `Manager.multiMap.GetTileLayerLookup` which gives you a job-compatible struct which can be used to access the same functions.

0.7.3

- You can choose a dedicated server install when choosing an install location in the SDK. Mods that are loaded via the SDK will also be detected properly by any connecting client even if the client uses the mod from `mod.io`. There is no automatic install of any missing mod that isn't installed via `mod.io` though.
- We are retiring the `Translation` component in favor of the `LocalTransform` component introduced in ECS 1.0. The main motivation for this change is to improve performance, as the `Translation` and `LocalTransform` must currently be kept in sync by copying the values back and forth. The `Translation` component will still be available in the project in order not to break serialized worlds, but no runtime entities are expected to have this component. This is one of the most commonly used components, but the following steps made the upgrade process fairly pain-free when we upgraded our own code:
 1. Use the following case-sensitive find-and-replace patterns. This took care of the vast majority of our usages.
 - ``Translation translation` -> `LocalTransform transform``
 - ``translation.Value` -> `transform.Position``
 - `` , translation) in` -> ` , transform) in``
 - ``<Translation>` -> `<LocalTransform>``
 - ``typeof(Translation)` -> `typeof(LocalTransform)``
 2. Fix all resulting compiler errors. The most common cause is that the patterns don't catch variable names in some cases.
 3. Replace all usages of ``new Translation`` with calls to ``LocalTransform.FromPosition``.
 4. Find and replace any remaining usages of the `Translation` component.

Unity's official ECS 1.0 upgrade guide might provide useful information on the new `LocalTransform` component:

<https://docs.unity3d.com/Packages/com.unity.entities@1.0/manual/upgrade-guide.html#update-transforms-in-your-project>.

- Completely removed the `Rotation` component, as rotation of entities is not used anywhere in Core Keeper. Setting the `Rotation` field on `LocalTransform` also has no effect.
- `EntityMonoBehaviour` is the base class for all of our graphical objects, and over the years it has grown to an unmanageable size as it has to support all our objects' specific use cases. We are making a push to return to a more component-based approach. Our aim is to make it less daunting and error-prone to add new graphical objects to the game. For this update specifically, we have:
 1. Added support for multiple `IGraphicalObject` components on the same object. All such components **at the object root** are picked up and have their methods invoked as appropriate.

2. Added separate interfaces for components that only provide spawn/despawn behavior. Prefer using these when possible to avoid the overhead of empty `GraphicalUpdate` methods being called each frame.
 3. Added several new components that control aspects of graphical objects that were previously managed with settings on `EntityMonoBehaviour`. We encourage you to when possible use these components instead of the corresponding `EntityMonoBehaviour` settings. The old settings have not been removed, but they will not be developed further and may be removed at a future time. These components are listed below, for more information see their in-editor tooltips and how they are used on existing assets.
``SpriteSkinFromPaintColor``, ``SpriteGradientMapFromPaintColor``,
``SpriteVariationFromEntityDirection``, ``SpriteVariationFromEntityVariation``,
``SpriteSkinFromEntityAndSeason``, ``EnableEffectsForEntityVariation``,
``OffsetFromEntityDirectionOrVariation``
- The hierarchies of a lot of our graphical objects have also had an overhaul to improve performance. In most cases this means porting an older object to use our in-house sprite rendering solution (`SpriteObject`). Apart from this we have also done lots of small changes to increase performance in various ways, see the performance section of the 0.7.3 changelog for the Core Keeper application for more information on this.

0.7.2

- Upgraded Unity version to 2022.3.10f1.
- Upgraded DOTS/ECS to 1.0. See links for upgrade instructions. The Translation component should still work as before, and any baking changes aren't relevant because we are using our own baking system (`PugConversion`).
 - <https://docs.unity3d.com/Packages/com.unity.entities@1.0/manual/upgrade-guide.html>
 - <https://docs.unity3d.com/Packages/com.unity.netcode@1.0/manual/upgrade-guide.html>
 - <https://docs.unity3d.com/Packages/com.unity.physics@1.0/manual/upgrade-guide.html>
- Burst version upgraded to 1.8.9.
- A lot of ECS systems have been converted to use `ISystem` instead of `SystemBase` to improve performance.
- Added an option for you to bypass the security check, adding a notice about it to the user: "Caution: 'Elevated Access' mods have increased access to resources outside the game like user files and internet. For the best experience, install only from reputable sources." The reflection API might be removed in the future in favor of using this option instead.
- The `Unity.Collections` package is embedded in the SDK, but the only change is that we have added the old `FixedString32/64` types which are still used for some serialized components. Changing these to the new versions would have broken serialization.
- The DOTS version uses a very different serialization format for ECS worlds. As a result there is a separate application in `CoreKeeper_Data/StreamingAssets/Patcher`

which is used to convert saves from before 0.7.2 to an interim format that the new version can read.